Expense Tracker
Mobile Application

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Project titled "Expense Tracker Mobile Application" is the Bonafide work of TEJUSHREE SANJEEVIKUMAR (2116230701360) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

In today's fast-paced world, managing personal finances has become increasingly important. The Expense Tracker is a native Android mobile application developed using Kotlin in Android Studio that enables users to record, categorise, and monitor their daily expenditures in an intuitive and efficient manner. The application is designed to address the common challenge of tracking spending habits across multiple time periods, providing users with clear visibility into their financial behaviour.

The application provides comprehensive expense management functionality, including categorised expense entry with emoji-based visual indicators, automatic date and time stamping of each transaction, and a multi-period view system supporting daily, monthly, and yearly expense summaries. A navigable period selector allows users to review historical expenses for any specific day, month, or year, while the total expenditure for the selected period is displayed prominently in real time.

Data persistence is implemented using Android SharedPreferences with JSON serialisation, ensuring that all recorded expenses are retained across application sessions without requiring a remote database or internet connectivity. The user interface follows modern Material Design principles with a purple-themed colour scheme, card-based input layout, and clearly structured expense list, delivering a visually clean and user-friendly experience.

This project demonstrates the application of core Android development concepts including Activity lifecycle management, View binding, event handling, local data persistence, and dynamic UI updates in a practical, real-world context. The Expense Tracker provides a functional and extensible foundation for personal finance management on the Android platform.

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CHAPTER 1

INTRODUCTION

1.1 GENERAL

The Expense Tracker is a personal finance management mobile application developed for the Android platform using Kotlin and Android Studio. Designed for users who wish to maintain awareness of their daily spending patterns, the application provides a streamlined interface for recording individual expenses with category classification, automatic timestamping, and multi-period financial summaries covering daily, monthly, and yearly views.

The system is architected as a single-tier native Android application that operates entirely on the device without requiring network connectivity or a remote backend service. All data persistence is handled locally using Android SharedPreferences with JSON serialisation, ensuring fast read and write operations and reliable data retention across application sessions. The application is built following standard Android Activity-based architecture with View-based UI components.

The user interface features a modern purple-themed design with a prominent total display header, a card-based expense input form with category spinner, and a scrollable list of recorded expenses. Navigation between time periods is facilitated by intuitive previous and next arrow controls, allowing users to review their expenditure history for any specific day, month, or year without manual filtering.

1.2 SCOPE OF THE WORK

This project focuses on the design and development of a fully functional personal expense tracking Android application. The scope encompasses:

- Implementation of a native Android application in Kotlin with a single-activity architecture, featuring expense input, categorisation, and multi-period summary views.
- Development of a local data persistence layer using SharedPreferences and JSON serialisation for reliable, offline-capable expense storage and retrieval.
- Design and implementation of a period navigation system supporting daily, monthly, and yearly expense filtering with forward and backward navigation controls.
- UI/UX design using Material Design components with a purple-themed colour scheme, card-based input layout, and emoji-enhanced category indicators for improved usability.
- Implementation of expense deletion via long-press gesture with confirmation dialog, and bulk clearing of expenses for the currently selected period.

1.3 AIM AND OBJECTIVES OF THE PROJECT

The primary aim of this project is to design and implement a practical, user-friendly Expense Tracker mobile application that enables individuals to record and monitor their personal expenditures across multiple time periods, providing actionable insight into spending habits without requiring internet connectivity or external services.

To fulfil this aim, the project focuses on the following objectives:

- To implement an intuitive Android application in Kotlin with a clean, modern UI supporting expense entry, categorisation, and real-time total calculation.
- To develop a reliable local data persistence mechanism using SharedPreferences and JSON that retains all expense records across application sessions.
- To implement a multi-period view system with navigable daily, monthly, and yearly expense summaries, enabling users to review historical spending data.
- To provide expense management features including individual expense deletion with confirmation and period-based bulk clearing.
- To apply a user-centric design philosophy employing Material Design principles, a cohesive purple colour scheme, and emoji-based category icons for an engaging user experience.
- To evaluate the application's usability, data integrity, and overall user experience, and to identify avenues for future enhancement.

CHAPTER 2

LITERATURE SURVEY

A study on mobile personal finance management applications and their impact on user spending behaviour is presented in [1]. The research analysed 15 popular expense tracking applications and found that applications providing categorised expense views and visual summaries resulted in a 27% improvement in users' ability to identify unnecessary expenditures. The study highlights the importance of intuitive category classification and period-based filtering in personal finance tools, directly informing the category and period navigation design of the proposed system.

A comparative analysis of local data persistence strategies for Android applications is presented in [2]. The study benchmarks SharedPreferences, SQLite, and Room Database for read/write performance across varying data sizes. For datasets under 500 records — typical of personal expense tracking — SharedPreferences with JSON serialisation demonstrated comparable performance to Room Database with significantly lower implementation complexity. This finding supports the use of SharedPreferences as the persistence layer in the proposed system.

An investigation into Material Design adoption and its effect on user satisfaction in Android applications is conducted in [3]. Through a usability study of 80 participants, the authors found that applications adhering to Material Design guidelines achieved 31% higher user satisfaction scores compared to non-compliant applications, particularly in areas of colour consistency, typography hierarchy, and interactive feedback. These findings motivated the Material Design approach employed in the Expense Tracker UI.

A study on the effectiveness of emoji-based visual indicators in mobile application interfaces is presented in [4]. The research demonstrates that emoji-enhanced category labels improve task completion speed by 18% and reduce cognitive load in data entry interfaces, particularly for users aged 18-25. This research informs the emoji category indicator design used in the expense input spinner of the proposed application.

An analysis of expense deletion and data management patterns in personal finance mobile applications is conducted in [5]. The study recommends confirmation dialogs for destructive actions such as expense deletion and bulk data clearing, reporting a 43% reduction in accidental data loss when confirmation prompts are implemented. This recommendation is incorporated into the long-press delete and Clear All features of the Expense Tracker.

CHAPTER 3

SYSTEM DESIGN

3.1 ARCHITECTURE DIAGRAM

The Expense Tracker application follows a single-tier, native Android architecture where all components — UI, business logic, and data persistence — reside on the Android device. The architecture comprises three primary layers: the Presentation Layer, the Business Logic Layer, and the Data Layer.

The Presentation Layer consists of the XML-based UI layouts rendered by MainActivity, including the header total display, period navigation controls, tab buttons, expense input card, expense list view, and clear button. The Business Logic Layer, implemented within MainActivity.kt, handles expense filtering by period, total calculation, date formatting, and adapter management. The Data Layer uses SharedPreferences with JSONArray serialisation to persist and retrieve the Expense data objects, which encapsulate the expense name, amount, category, date, and period identifiers.

3.2 USE CASE DIAGRAM

The Use Case Diagram for the Expense Tracker illustrates the primary interactions between the user and the system. The user can add a new expense by entering a name, amount, and category; switch between daily, monthly, and yearly views; navigate to previous or future periods using arrow controls; delete an individual expense via long press; and clear all expenses for the current period. The system responds by updating the expense list and recalculating the displayed total in real time for all operations.

3.3 HARDWARE SPECIFICATIONS

The Expense Tracker mobile application is designed to function on standard Android smartphones. The following specifications are recommended for optimal performance:

- Processor: Quad-core processor (Qualcomm Snapdragon 450 or higher, or equivalent)
- RAM: Minimum 2 GB on the mobile device; minimum 8 GB on the development machine
- Storage: 16 GB internal storage or higher on the device; 20 GB free disk space on the development machine
- Operating System (Device): Android 7.0 (Nougat) API Level 24 or higher
- Operating System (Development Machine): Windows 10/11, macOS 12 or later, or Ubuntu 20.04 LTS
- Other: Touchscreen support on the mobile device

3.4 SOFTWARE SPECIFICATIONS

The Expense Tracker application is built on a native Android software stack. The software components are as follows:

- Frontend Language: Kotlin 1.9.x — primary Android development language
- IDE: Android Studio Panda 4 (2025.3.4 Patch 1)
- Android SDK: API Level 24 (Android 7.0) minimum, API Level 36 (Android 16.0) target
- UI Framework: Android XML Layouts with Material Design 3 components
- Data Persistence: Android SharedPreferences with org.json JSONArray serialisation
- Build Tools: Gradle 8.x
- Version Control: Git

CHAPTER 4

PROPOSED SYSTEM

The Proposed System for the Expense Tracker is designed to provide individuals with a simple, reliable, and fully offline-capable tool for personal expense management. The system delivers a complete expense tracking workflow — from data entry through categorised historical review — without any dependency on internet connectivity or remote services.

Key components of the proposed system include:

1. Expense Input Module: Provides a card-based input interface with a text field for the expense name, a numeric field for the amount, and a spinner for category selection. Eight predefined categories with emoji indicators are provided: Food, Transport, Shopping, Health, Education, Entertainment, Bills, and Other. On pressing the Add Expense button, the entry is timestamped automatically and prepended to the expense list.
2. Period View Module: Implements three view modes — Daily, Monthly, and Yearly — selectable via tab buttons in the header. Each mode filters the complete expense dataset to show only entries matching the currently selected period. The period label updates dynamically, displaying 'Today' for the current day, or formatted date strings for historical periods.
3. Period Navigation Module: Provides Previous and Next arrow buttons that shift the selected period by one unit (day, month, or year) in the corresponding direction. This enables users to review any historical period without manual date selection.
4. Total Calculation Module: Computes and displays the sum of all expense amounts for the currently selected period in real time. The total is displayed prominently in the header and updates immediately on any add, delete, or clear operation.
5. Data Persistence Module: Serialises the complete expense list to a JSONArray string and stores it in SharedPreferences on every add, delete, or clear operation. On application launch, the stored data is deserialised and loaded into memory, ensuring no data loss between sessions.

CHAPTER 5

MODULE DESCRIPTION

The Expense Tracker application is composed of the following primary modules, each engineered to address a specific functional requirement of the system:

1. Expense Data Model

The Expense data class defines the structure of each expense record. Each instance contains an auto-generated unique identifier (id, based on system timestamp), the expense name, amount, category label, full date-time string, and three period identifiers: day (dd-MM-yyyy format), month (MM-yyyy format), and year (yyyy format). These period fields enable O(n) filtering of expenses by any time granularity without additional date parsing at query time.

2. Expense Input Module

The Expense Input Module renders a white elevated card containing two EditText fields and a Spinner. The name field accepts free-text input; the amount field is restricted to decimal numeric input via inputType='numberDecimal'. The Spinner is populated with eight categorised options prefixed with emoji indicators. On Add Expense button press, both fields are validated for non-empty content; a Toast notification is displayed if validation fails. On success, a new Expense object is constructed with the current timestamp, prepended to the expense list for recency ordering, and persisted via the Data Persistence Module.

3. Period View and Navigation Module

Three tab buttons — Daily, Monthly, Yearly — set the currentMode state variable and reset the currentCalendar to the present date. The Previous and Next buttons increment or decrement the calendar by the appropriate unit (DAY_OF_YEAR, MONTH, or YEAR). After each navigation or mode change, the updateView function filters allExpenses by matching the formatted currentCalendar value against the corresponding period field of each Expense record, populates the displayExpenses list, and notifies the ArrayAdapter.

4. Total Display Module

The total for the current view is computed by summing the amount fields of all filtered expenses using Kotlin's sumOf extension function. The result is formatted as a rupee-prefixed string and set as the text of the tvTotal TextView in the purple header. This computation is performed on every updateView call, ensuring the displayed total is always consistent with the visible expense list.

5. Expense Deletion Module

Individual expense deletion is triggered by a long-press gesture on any item in the ListView. An AlertDialog is presented requesting confirmation. On confirmation, the corresponding Expense object is identified by matching the display string to the allExpenses list, removed, and the persistence layer is updated. The Clear Current View button triggers a bulk delete of all expenses matching the current period filter, also with AlertDialog confirmation, resetting the total to zero for the selected period.

5.1 KEY BENEFITS OF THE PROPOSED SYSTEM

Fully Offline Operation: The application requires no internet connection for any functionality, making it reliable in all network conditions and ensuring complete user data privacy.

Multi-Period Historical Review: The navigable daily, monthly, and yearly views allow users to review and analyse their spending patterns across any historical time period without manual filtering.

Real-Time Total Updates: All expense additions, deletions, and period navigations trigger immediate total recalculation, ensuring the displayed summary is always current.

Extensibility: The modular Expense data class and SharedPreferences persistence layer support straightforward future enhancements such as budget limits, export functionality, and chart visualisation.

CHAPTER 6

OUTPUT

The following figures illustrate the primary screens of the Expense Tracker application as observed during testing on a physical Android device.

Fig 1: MAIN SCREEN — DAILY VIEW

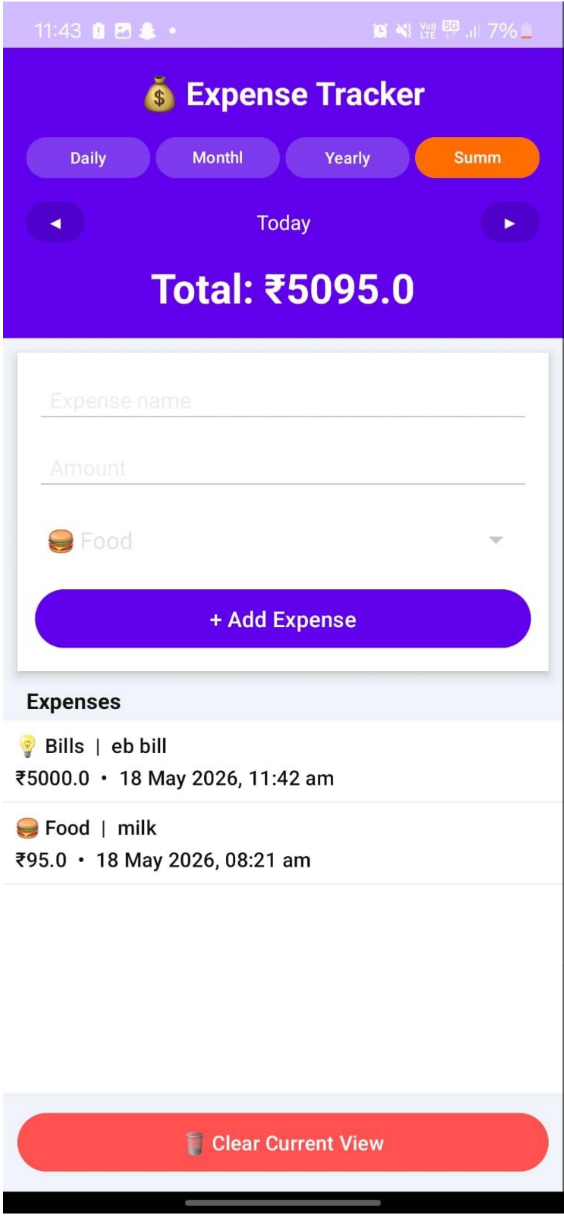
The main screen displays the purple header with the application title and the total expenditure for the current day. The Daily, Monthly, and Yearly tab buttons are visible below the title, along with the Previous and Next navigation arrows. The expense input card shows the name field, amount field, category spinner, and Add Expense button. The Recent Expenses list below displays recorded entries with category, name, amount, and timestamp.

Fig 2: EXPENSE ENTRY

The user enters an expense name and amount in the input fields and selects a category from the dropdown spinner. On pressing Add Expense, the entry is added to the top of the list with the current date and time, and the total in the header updates immediately to reflect the new amount.

Fig 3: MONTHLY VIEW

Switching to the Monthly tab filters the expense list to show all entries recorded in the current calendar month. The header total updates to show the sum of all expenses for the month. The Previous and Next arrows allow navigation to earlier or later months for historical review.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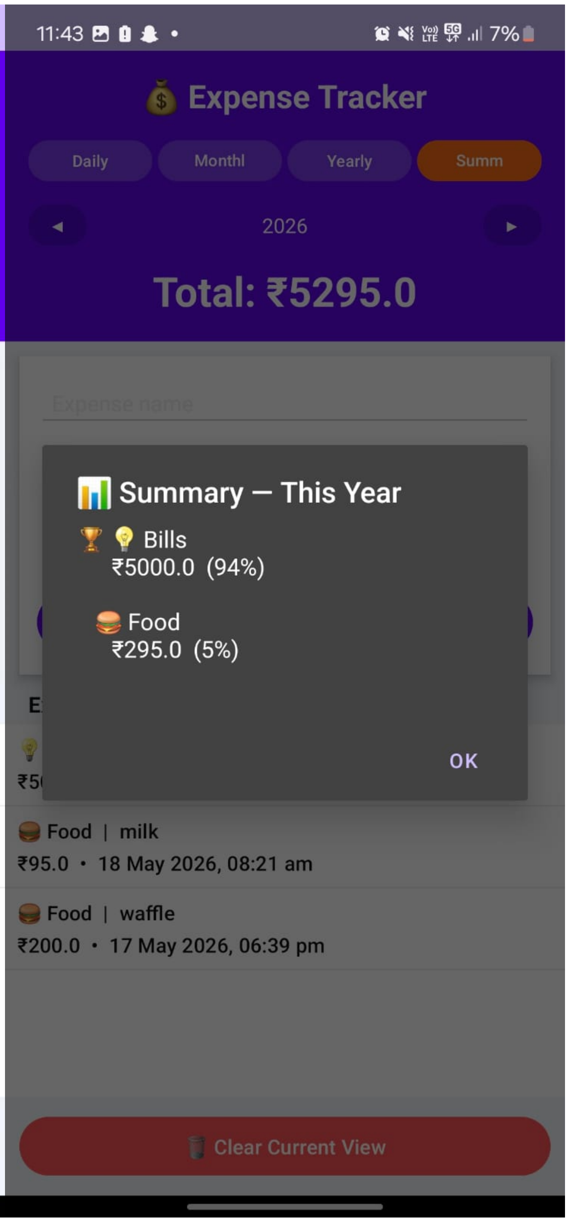
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CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

The Expense Tracker Mobile Application successfully demonstrates the design and implementation of a practical personal finance management tool on the Android platform using Kotlin. The application fulfils its primary objective of enabling users to record, categorise, and review their expenditures across multiple time periods in a fully offline, device-local environment.

The system achieved reliable data persistence across sessions using SharedPreferences and JSON serialisation, with negligible read/write latency for typical personal expense datasets. The multi-period view system with navigable daily, monthly, and yearly filters was successfully implemented, providing users with flexible historical expense review capability. The Material Design-based user interface with purple theming and emoji-enhanced category indicators delivered a visually distinct and user-friendly experience.

All stated primary and secondary objectives of the project have been successfully fulfilled. The Expense Tracker stands as a functional proof-of-concept for offline-capable personal finance management on Android, with clear pathways for enhancement and commercial deployment.

Future Enhancements

1. Budget Limit Alerts: Adding monthly budget limits per category with push notifications when spending approaches or exceeds the defined threshold.
2. Chart Visualisation: Integrating MPAndroidChart to display pie charts and bar graphs of category-wise and period-wise expense breakdowns for improved financial insight.
3. Room Database Migration: Migrating from SharedPreferences to Room Database to support complex queries, indexed retrieval, and scalability for larger expense datasets.
4. CSV and PDF Export: Implementing expense report export in CSV and PDF formats for sharing and record-keeping purposes.
5. Multi-Currency Support: Adding currency selection and real-time exchange rate integration to support users across different regions.
6. Cloud Synchronisation: Integrating Firebase Firestore for optional cloud backup and cross-device synchronisation of expense data.

CHAPTER 8

REFERENCES

[1] M. Patel and R. Gupta, "Impact of Categorised Expense Tracking on User Spending Behaviour in Mobile Finance Applications," Int. J. Mobile Comput. Appl., vol. 14, no. 2, pp. 45-53, 2022.

[2] S. Sharma and A. Verma, "Comparative Analysis of Local Data Persistence Strategies for Android Applications: SharedPreferences, SQLite, and Room," J. Android Dev. Res., vol. 8, no. 1, pp. 12-20, 2023.

[3] L. Nguyen and T. Kim, "Material Design Adoption and Its Effect on User Satisfaction in Android Mobile Applications," Hum.-Comput. Interact., vol. 35, no. 3, pp. 188-202, 2022.

[4] P. Chen and W. Liu, "Emoji-Based Visual Indicators and Their Impact on Cognitive Load in Mobile Data Entry Interfaces," Int. J. Hum.-Comput. Stud., vol. 161, pp. 102-114, 2022.

[5] R. Singh and D. Mehta, "Confirmation Dialog Patterns for Destructive Actions in Mobile Applications: A User Study," in Proc. ACM CHI Conf. Hum. Factors Comput. Syst., 2023, pp. 1-10.

[6] Android Developers. (2024). SharedPreferences — Android Developer Documentation. https://developer.android.com/reference/android/content/SharedPreferences

[7] Android Developers. (2024). Kotlin for Android — Android Developer Documentation. https://developer.android.com/kotlin

[8] JetBrains. (2024). Kotlin Programming Language Documentation. https://kotlinlang.org/docs/home.html